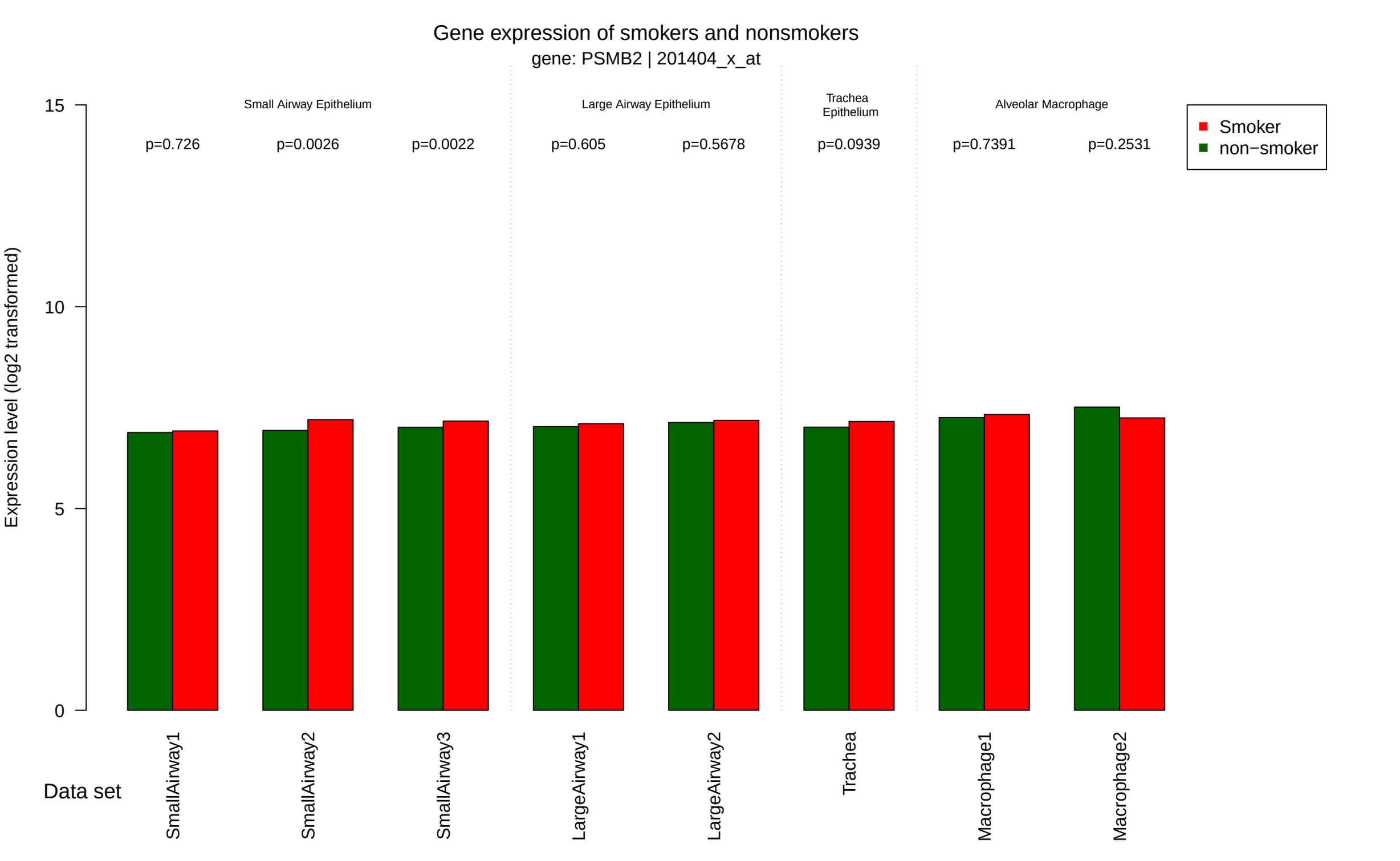
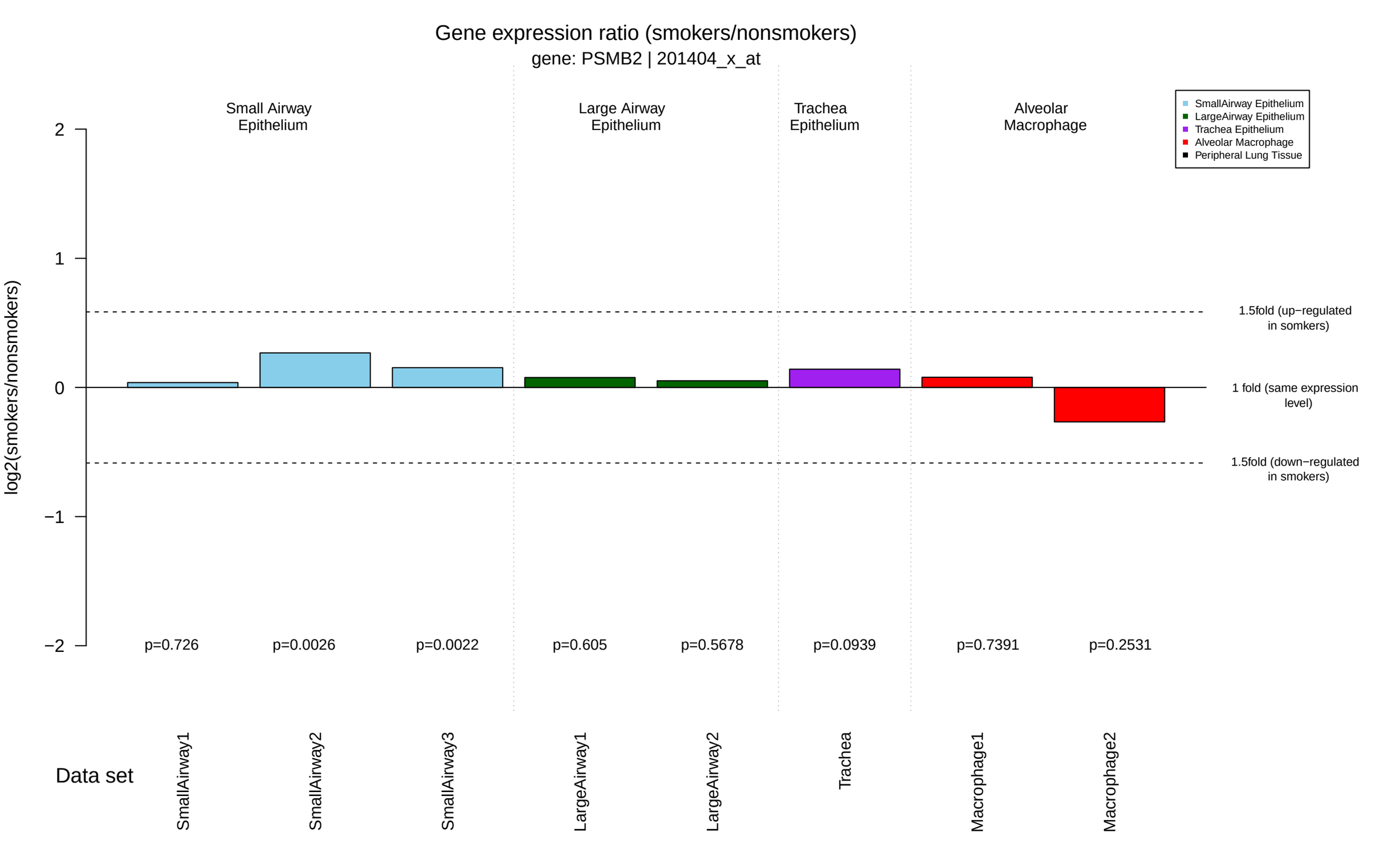
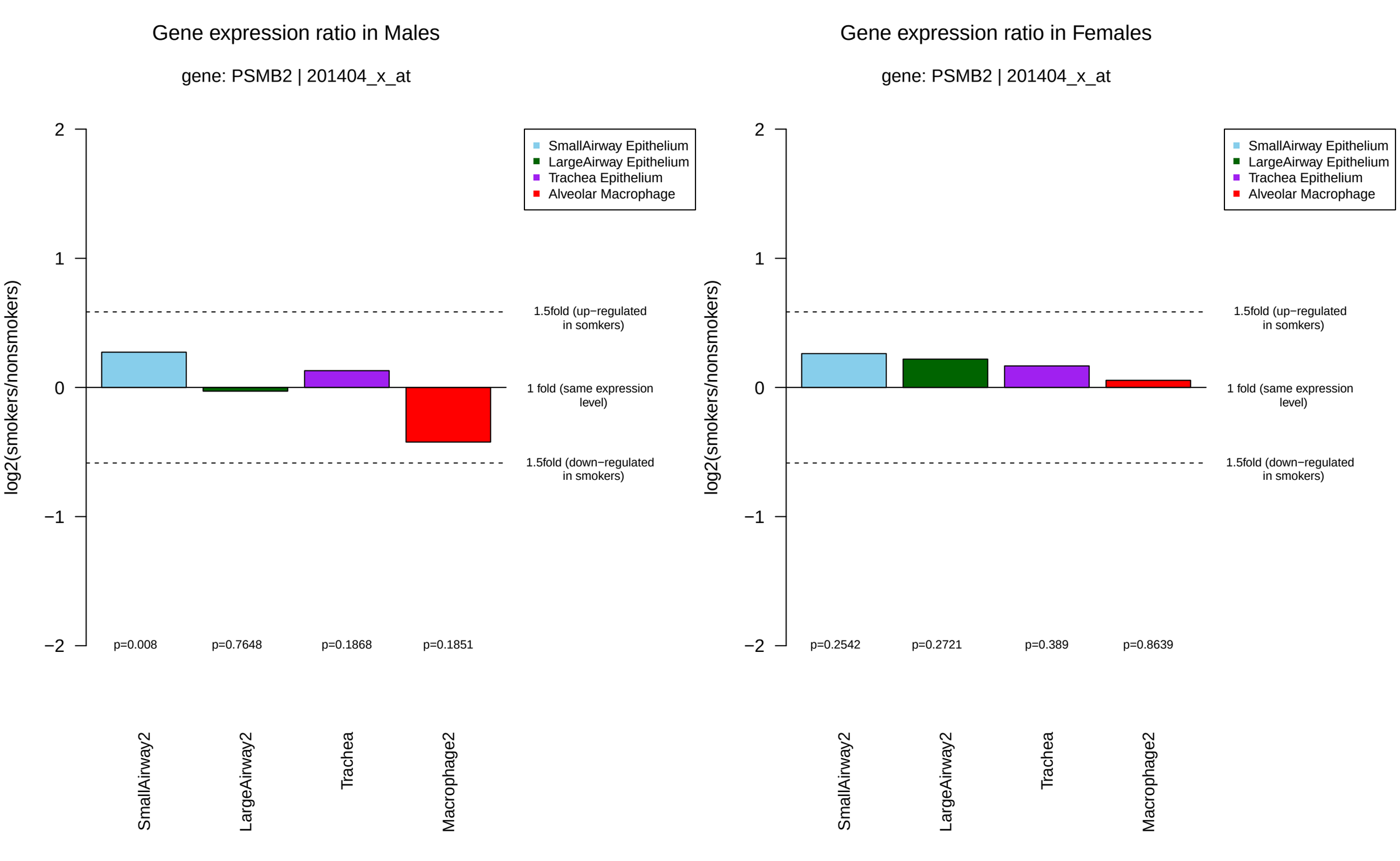
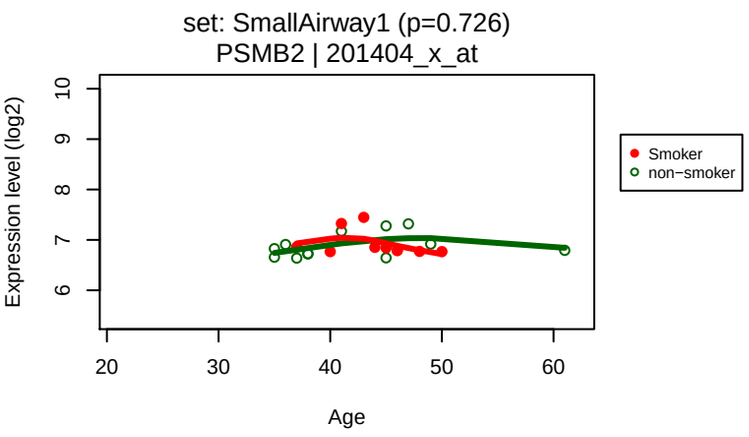
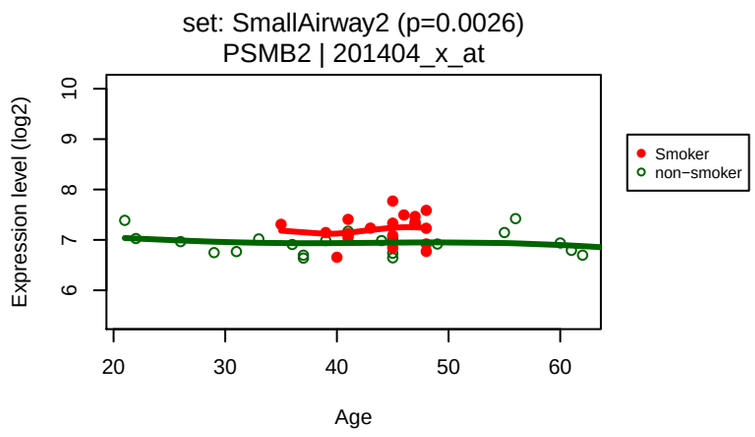
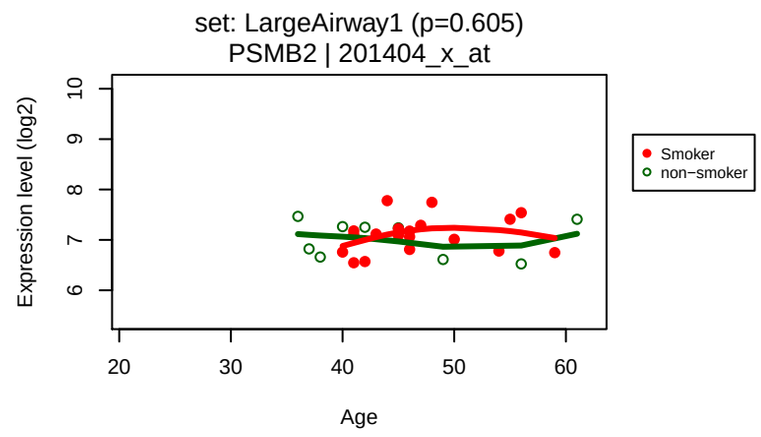

Gene expression of smokers and nonsmokers
gene: PSMB2 | 201404_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.726
p=0.0026
p=0.0022
p=0.605
p=0.5678
p=0.0939
p=0.7391
p=0.2531
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

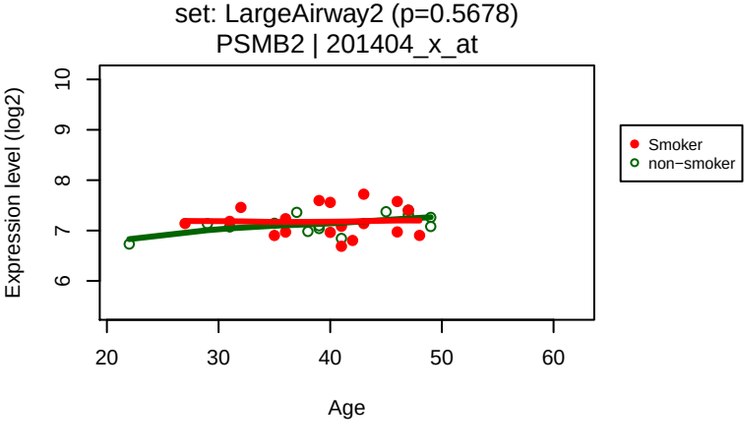

Gene expression ratio (smokers/nonsmokers)
gene: PSMB2 | 201404_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.726
p=0.0026
p=0.0022
p=0.605
p=0.5678
p=0.0939
p=0.7391
p=0.2531
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

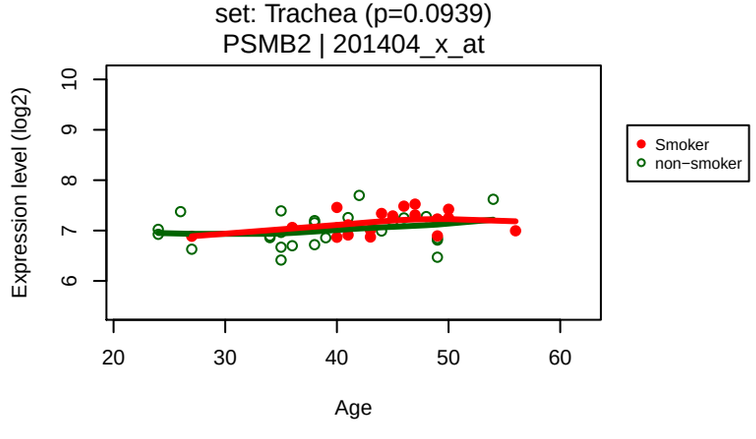

Gene expression ratio in Males
gene: PSMB2 | 201404_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.008
p=0.7648
p=0.1868
p=0.1851
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB2 | 201404_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2542
p=0.2721
p=0.389
p=0.8639
SmallAirway2
LargeAirway2
Trachea
Macrophage2

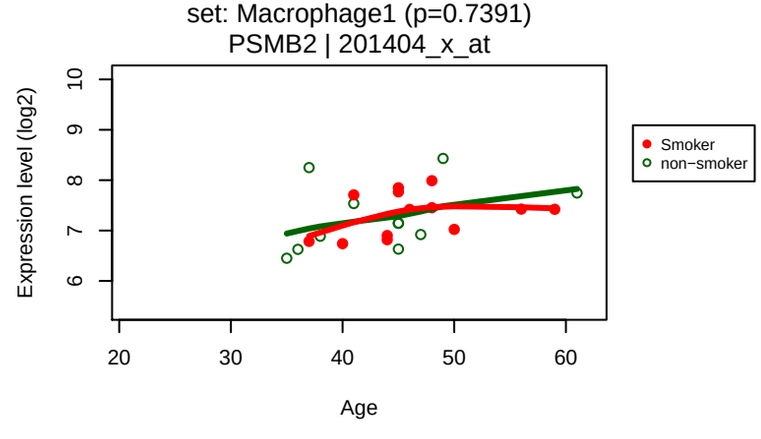

set: SmallAirway1 (p=0.726)
PSMB2 | 201404_x_at
Expression level (log2)
Age
Smoker
non−smoker

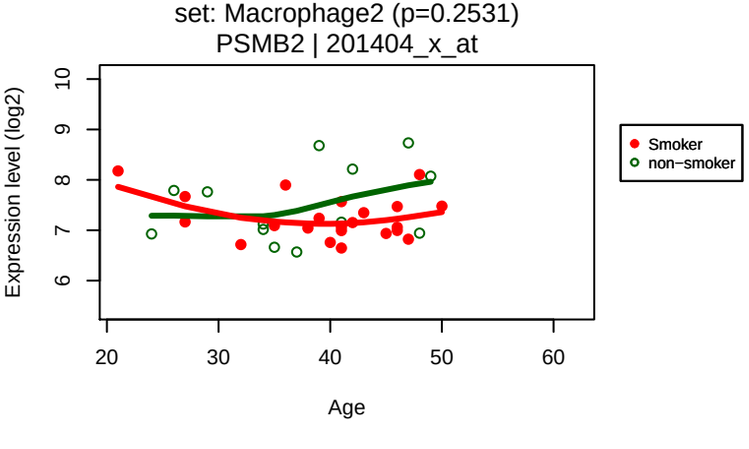

set: SmallAirway2 (p=0.0026)
PSMB2 | 201404_x_at
Expression level (log2)
Age
Smoker
non−smoker

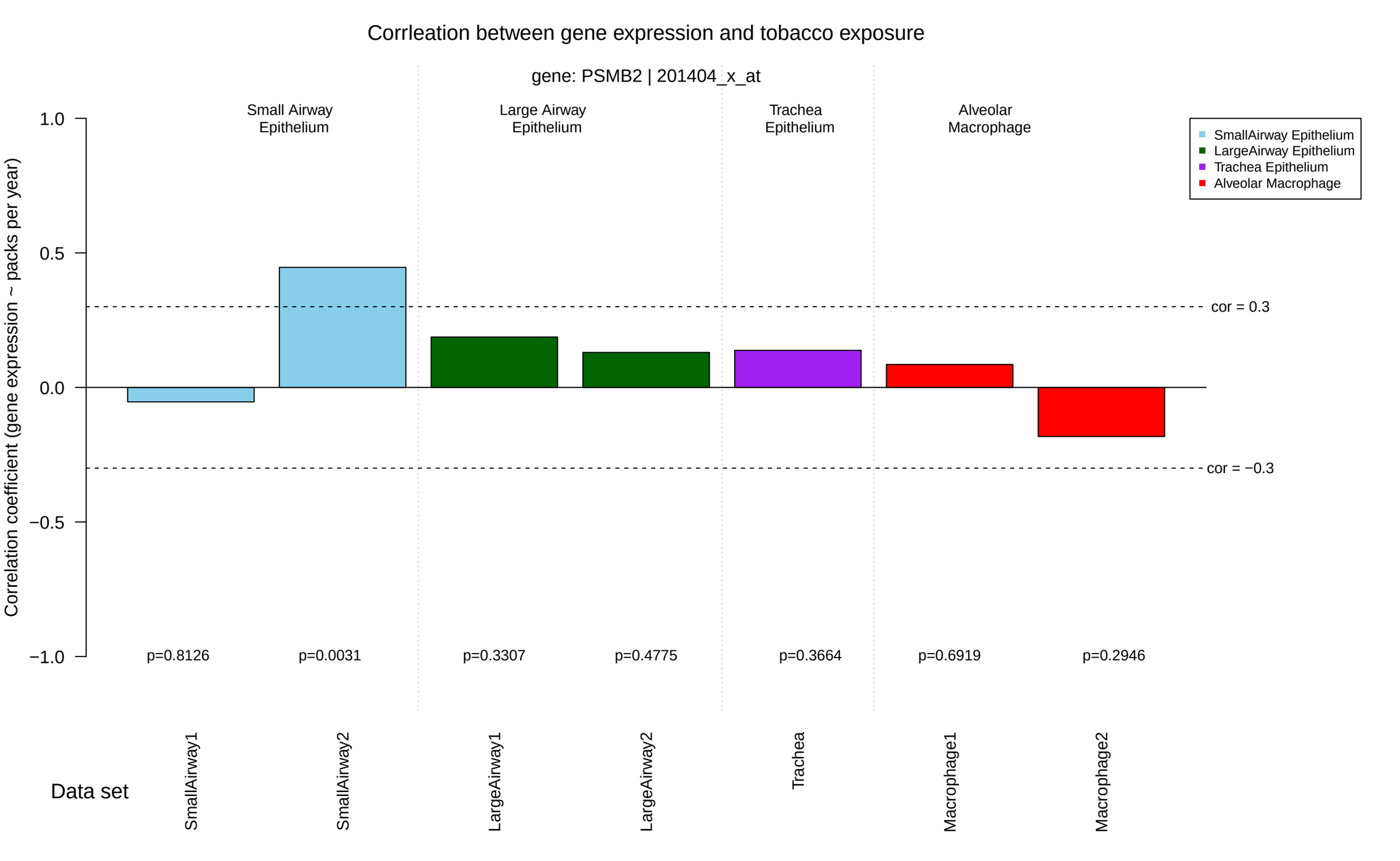

set: LargeAirway1 (p=0.605)
PSMB2 | 201404_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5678)
PSMB2 | 201404_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0939)
PSMB2 | 201404_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7391)
PSMB2 | 201404_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2531)
PSMB2 | 201404_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB2 | 201404_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8126
p=0.0031
p=0.3307
p=0.4775
p=0.3664
p=0.6919
p=0.2946
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2